Le Châtelier’s Principle

Chem 151 Lab

(Qualitative Lab – make sure to record all observations)

Chemical Equilibrium

- Equilibrium - state in which the forward and reverse reactions proceed at the same rate.
- Concentrations of reactants and products remain the same, once equilibrium is reached.
- The equilibrium state is *dynamic* – both forward and reverse reactions are still occurring, the reaction just *appears* static because the **CONCENTRATIONS of reactants and products are CONSTANT**.

General Reaction:



$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Solids and pure liquids (solvents) do not appear in equilibrium expressions - concentrations do not change!

Disturbing Systems at Equilibrium

<u>Le Chatelier's Principle</u> - a change in any of the factors that determine the equilibrium will cause the system to adapt in a manner to reduce the effect of that change

1. Adding/Removing a reactant or product
2. Changing the temperature
3. Changing the volume (or pressure) of the reaction vessel

   for rxns with unequal moles of gases in reactants and products a decrease in volume will cause a shift to the side with less moles of gas

Chromate – Dichromate equilibrium in aqueous solution

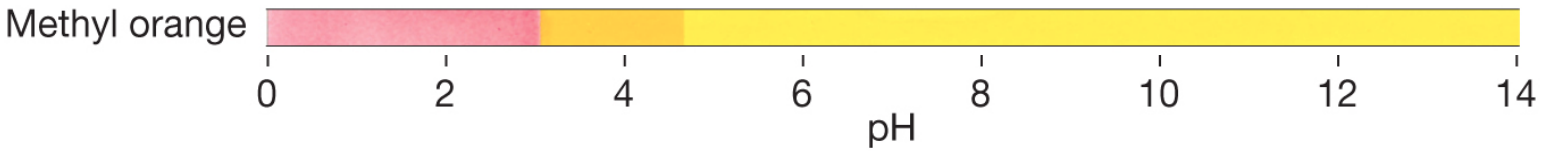
$$CrO_4^{2-}(aq) + H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)\ unbalanced$$

What happens when you add:

1. 3 M H_2SO_4?
2. 1 M NaOH?

Acetic Acid – Acetate ion equilibrium in aqueous solution

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$



What happens when you add:

1. Solid $NaCH_3COO$?

Tetrachlorocobaltate(II) – hexaquacobalt(II) equilibrium in aqueous solution.

$$CoCl_4^{2-}(aq) + 6H_2O(l) \rightleftharpoons Co(H_2O)_6^{2+}(aq) + 4Cl^-(aq)$$

What happens when you add:

1. Conc. HCl?
2. Distilled H_2O?

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Equilibrium solid ammonium chloride to ammonium and chloride ions in aqueous solution

Heat + $NH_4Cl(s) + H_2O(l) \rightleftharpoons NH_4^+(aq) + Cl^-(aq)$ ΔH_{rxn} = + 15.1 kJ/mol

Note: the states of matter!

What happens when you:

1. Add conc. HCl?
2. Heat up the system in equilibrium?

Hints for success:

- Make sure to record all observations
- Make sure to address **ALL** questions that are asked in the manual
- No conclusion is necessary for this week's lab
- Turn your notebook in by 11:59pm on Wednesday
- Feel free to use these slides as a guide